

Validation of the Accuracy of the OffAxisProjection Script

- Method

1. Use a script to obtain the world coordinates and dimensions of the selected objects in Unity.
2. Create three screens at appropriate positions in Unity so that their geometry matches the physical screens in the laboratory. Each screen is 4 m wide and 2.5 m high; the three screens are connected at 90° angles, and their lower edges are 0.2 m above the floor.
3. Determine an appropriate position for the eyeroot in Unity.
4. Using the perpendicular distance from the eyeroot to each screen and the distance from the eyeroot to the center point of the selected object, calculate the projection scale factor of the object on the corresponding screen. An object located in front of the screen plane appears enlarged, whereas an object located behind the screen plane appears reduced.
5. Select one object for each of the left, front, and right screens for calculation: the flowerpot on the left screen, the painting on the right screen, and the television on the front screen. The expected object size on the physical screen is calculated as: $\text{displayed size} = \text{projection scale factor} \times \text{object size in Unity}$.
6. The selected object should be approximately parallel to the corresponding screen.
7. Measure the displayed size of the object on the physical screen and compare it with the calculated result from Step 5.
8. The calculated and measured values differed by less than 5% in all cases. The remaining deviation may be caused by manual measurement errors and small offsets in the estimated object center point.